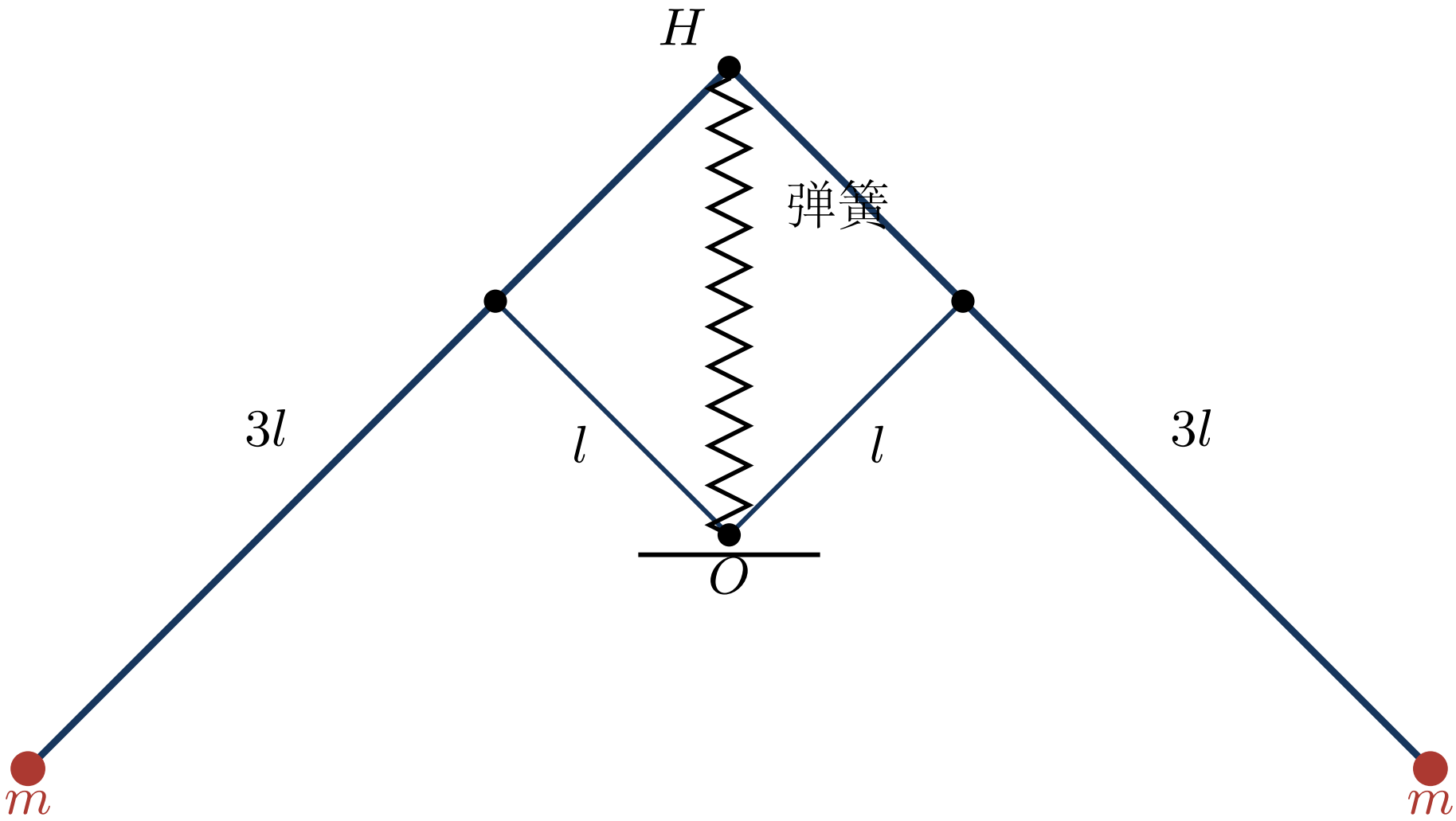
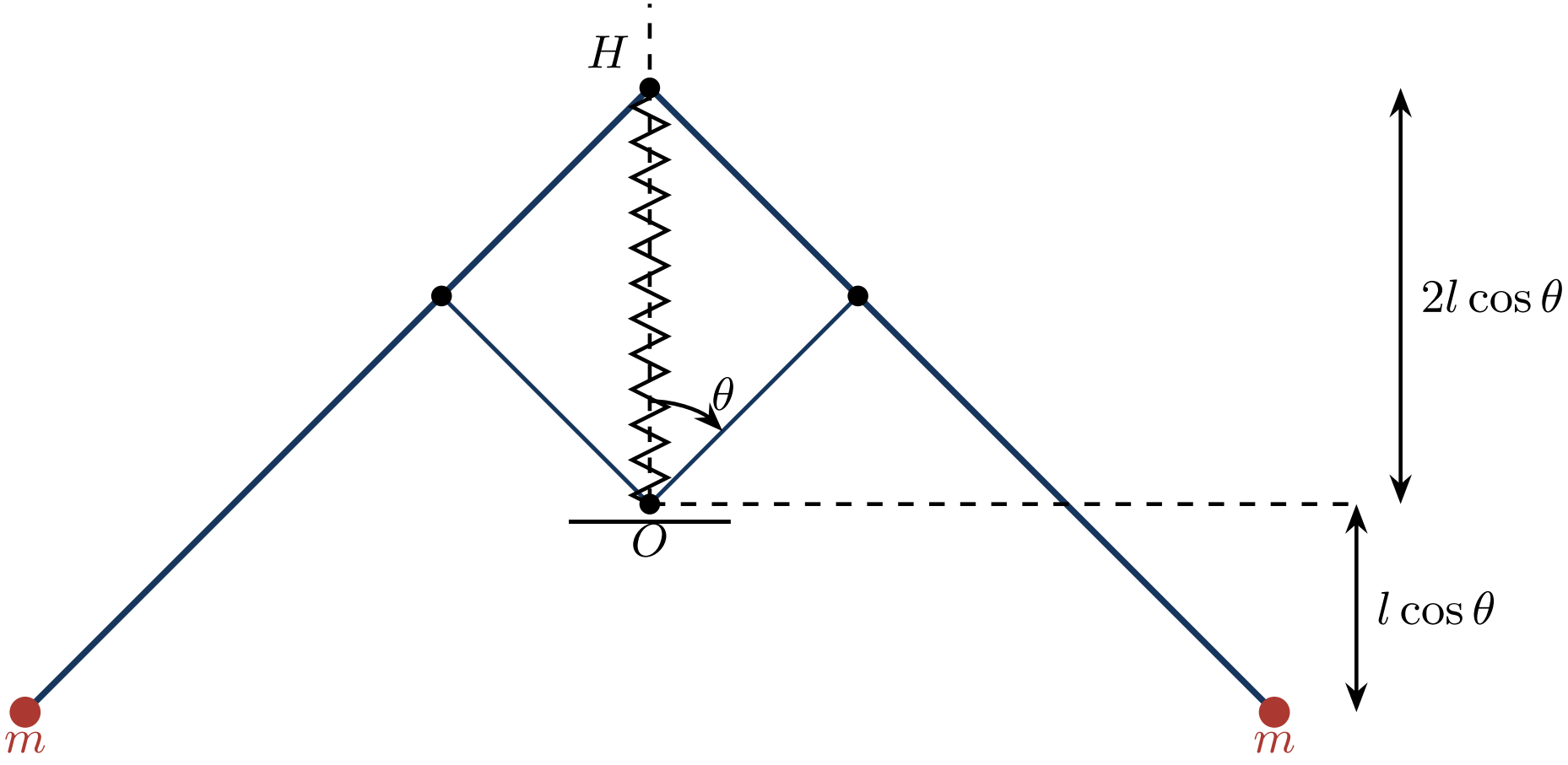
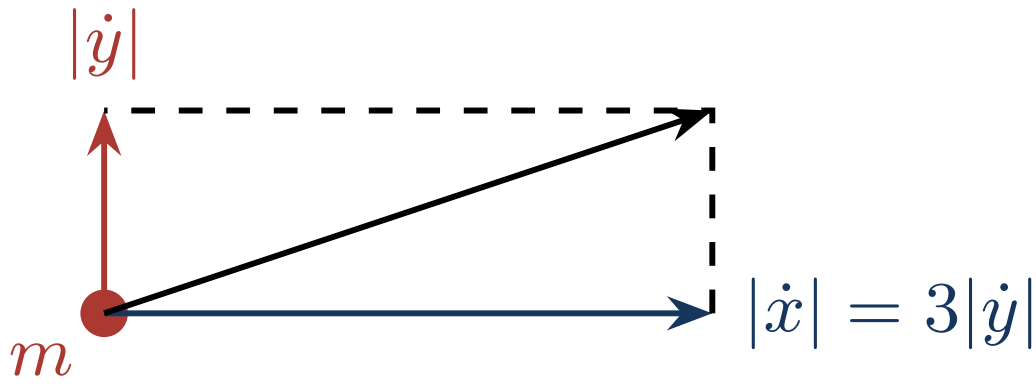


小菱形边长 l；长杆从 H 至外端长 $3l$。原长 $l_0 = l/\sqrt{2}$，平衡角为 45°。



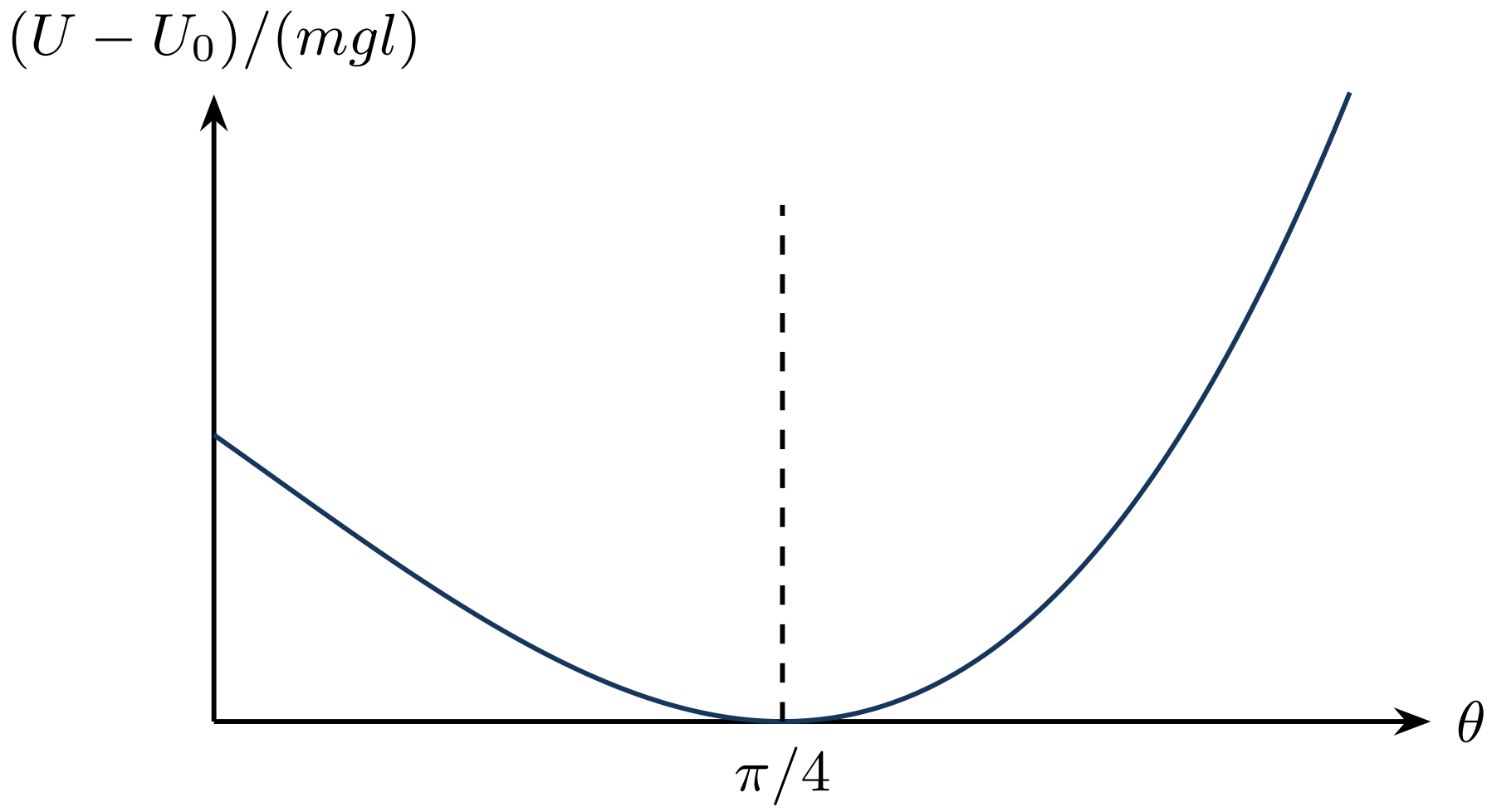
端点：$x = \pm 3l\sin\theta$，$z = -l\cos\theta$；弹簧长度 $L_{\mathrm{s}} = 2l\cos\theta$。

$y = l\cos\theta - l/\sqrt{2}$，向下为正；一级关系 $\dot{x} = -3\dot{y}$。

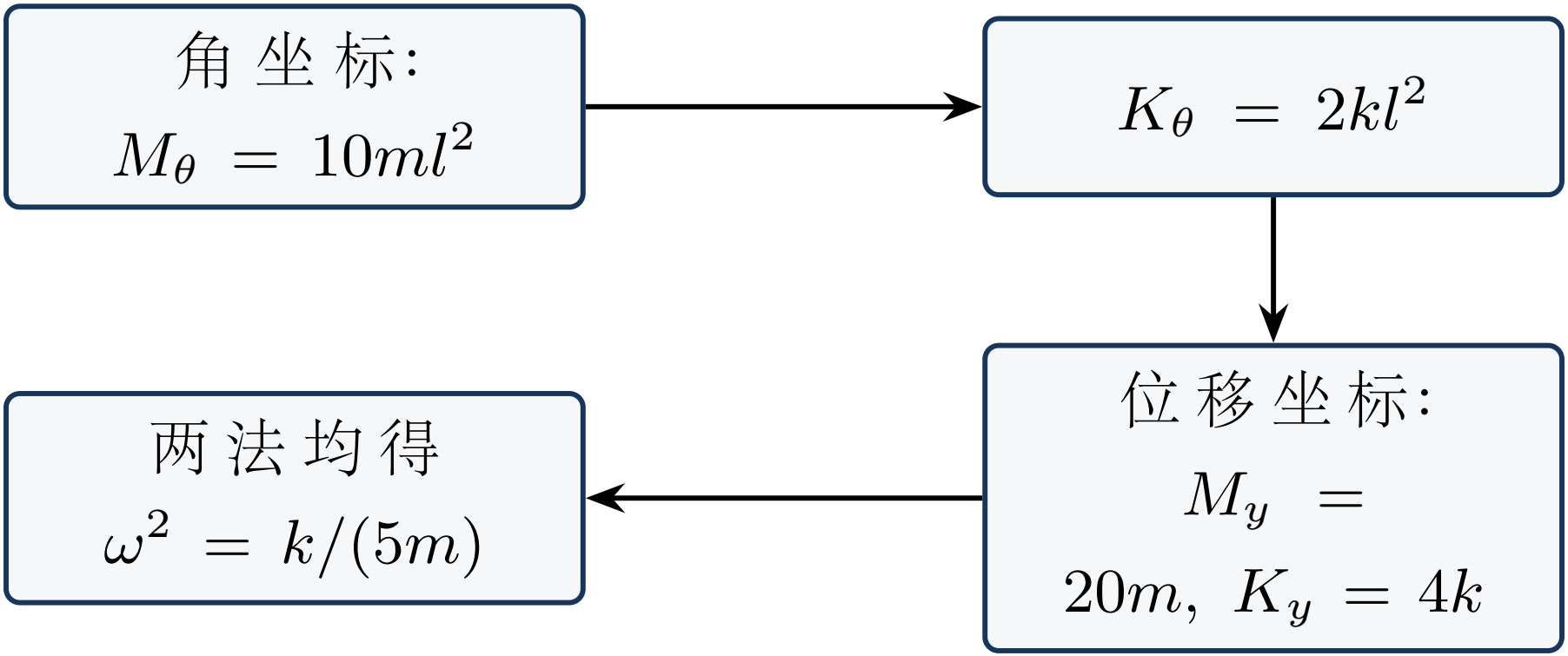


两质点总动能

$10m\dot{y}^2$



$U''(\pi/4) = 2kl^2 > 0$；图中能量自平衡值起算，横轴范围有限。



弹簧长度增量 $2y$; $U - U_0 = 2ky^2$。平衡条件给 $k = \sqrt{2}mg/l$。